

URL Shortener - URL Shortening Deep Dive

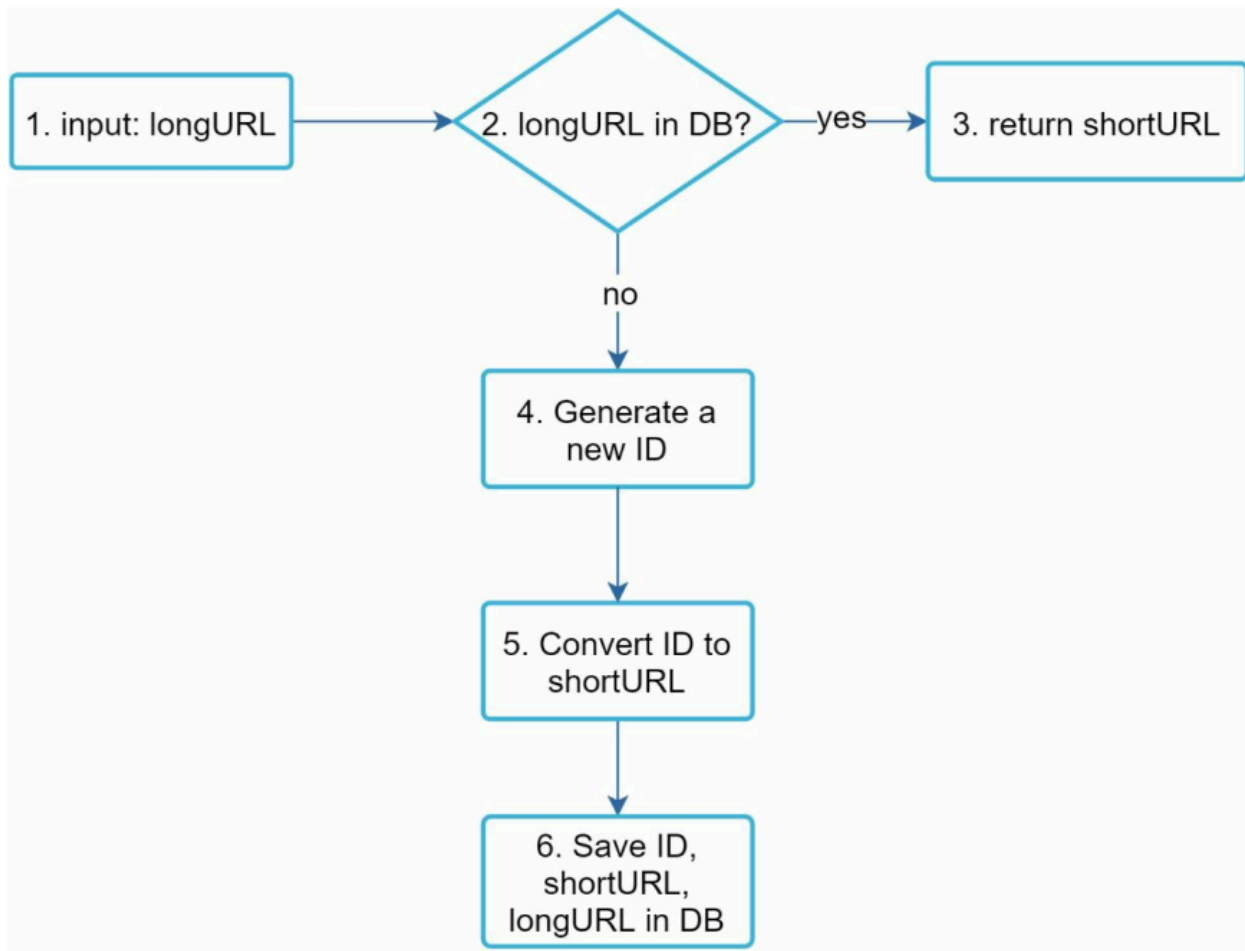
1. Overview

URL shortening is a **core component** of the system.

In this design:

- We use **Base62 conversion**
- Ensure:
 - Simplicity
 - Efficiency
 - Deterministic mapping

2. End-to-End Flow



Step-by-step process:

Step 1: Input

None

`longURL` is received from client

Step 2: Check Existing Mapping

None

Check if longURL exists in database

If exists:

- Fetch corresponding **shortURL**
- Return to client

Step 3: Handle New URL

If longURL is NOT found:

None

Proceed to generate new short URL

Step 4: Generate Unique ID

None

ID = Unique ID Generator

- Acts as **primary key**
- Must be:
 - Globally unique
 - Scalable

Step 5: Convert ID → Short URL

None

shortURL = Base62(ID)

Converts numeric ID into compact string

Step 6: Store in Database

Create new entry:

None

```
<ID, shortURL, longURL>
```

Step 7: Return Response

None

```
Return shortURL to client
```

3. Example Walkthrough

Input:

None

```
https://en.wikipedia.org/wiki/Systems_design
```

Step 1: Generate ID

None

```
ID = 2009215674938
```

Step 2: Base62 Conversion

None

```
shortURL = "zn9edcu"
```

Step 3: Store in Database

ID	shortURL	longURL
2009215674938	zn9edcu	https://en.wikipedia.org/wiki/Systems_design

Step 4: Return

None

<https://tinyurl.com/zn9edcu>

4. Important Component - Unique ID Generator

Role:

- Generate **globally unique IDs**
- Used as input for Base62 conversion

Requirements:

- High throughput
- Distributed support
- No collisions
- Time-ordered (optional but useful)

Note:

- Designing this component is **non-trivial**
- Covered separately in:

None

Chapter 7: Unique ID Generator (e.g., Snowflake)

5. Key Design Advantages

- No collision (ID is unique)
- Fast lookup (DB indexed by ID/shortURL)
- Reuse existing short URL for duplicate long URLs
- Scales well in distributed systems

6. Key Takeaways

- Base62 + Unique ID = efficient URL shortening
- Database ensures mapping persistence
- Duplicate URLs are handled via lookup
- ID generator is critical for scalability